

A

«CoSimComponent»

MultiComponent

+models: Dict[ModeKey, CoSimComponent]
+active_mode: ModeKey
+active_comp: Optional[CoSimComponent]
+mode_selector: Optional[Callable]
+hysteresis: Optional[Hysteresis]
+state_adapters: Dict[ModeKey, StateAdapter]

+__init__(name, initial_mode, group)

Abstract Methods

#_register_models(): None
#_adapt_state(state, target_mode): Dict[str, Any]

Initialization

#_initialize_component(t0)
-_unify_ports()

Mode Switching

#_do_step_internal(t, dt)
-_switch_mode(new_mode, t)

I/O Delegation

+set_inputs(signals, t)
#_update_output_states(t)

State Management

+set_state(state, t)
+get_state(): Dict[str, Any]
+reset()

C

Hysteresis

+dwell_time: float
+last_switch_time: float
+last_mode: ModeKey

+__init__(dwell_time)
+can_switch(t, proposed_mode): bool
+record_switch(t, new_mode)

uses

I

«Protocol»

StateAdapter

may use

+adapt_state(source_state, target_component): Dict[str, Any]

uses

C

«Str»

ModeKey

Mode identifier

e.g., "FEM", "OpenSim", "EQB"